IAT New Test 5

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

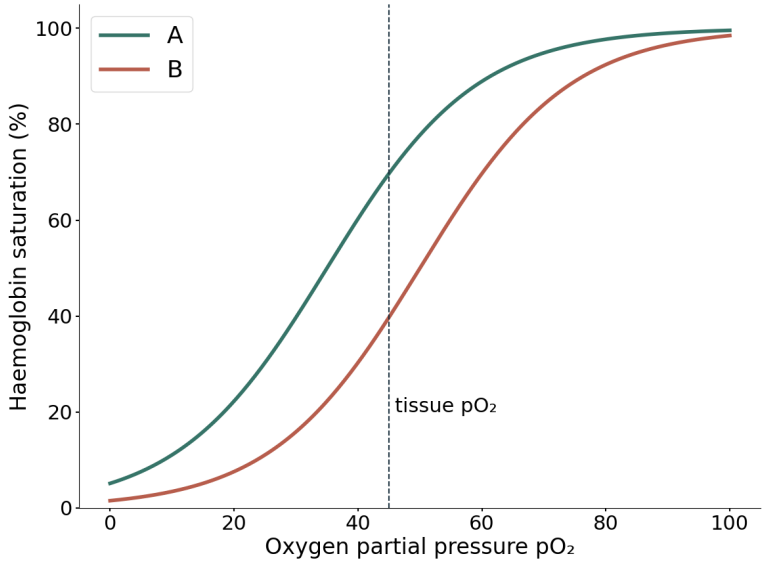
Q1 A seed develops from an ovule enclosed in an ovary. Mature seed has one cotyledon and parallel leaf venation. Which assignment combines the reproductive and vegetative evidence?
A) Monocot angiosperm
B) Dicot gymnosperm
C) Monocot gymnosperm
D) Dicot angiosperm

Q2 A frog's pulmonary veins are blocked while its systemic venous return remains open. Which statement correctly identifies the directly reduced inflow and a return route that is not directly blocked?
A) Left atrial inflow from skin; pulmonary return to right atrium remains
B) Left atrial inflow from lungs; systemic return through sinus venosus remains
C) Right atrial inflow from lungs; systemic return to left atrium remains
D) All atrial inflow stops immediately

Q3 In cell fractionation, fraction X is rich in catalase but poor in chlorophyll and mitochondrial cytochrome oxidase. Which organelle is enriched?
A) Chloroplast
B) Mitochondrion
C) Peroxisome
D) Nucleus

Q4 At constant light, sudden stomatal closure in a well-watered C_3 leaf most directly lowers which entry flux?
A) Mineral ions into roots
B) Sucrose into phloem
C) Oxygen into mitochondria only
D) CO_2 from air into intercellular spaces

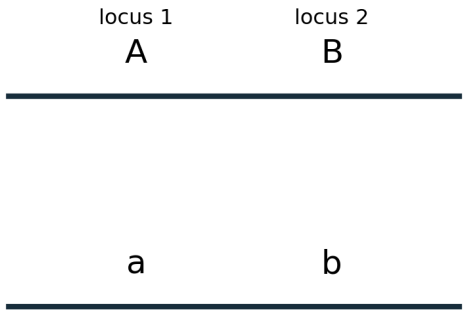
Q5 At a fixed tissue pO_2, curve B in the figure has lower haemoglobin saturation than curve A. If B is measured after an increase in CO_2 and temperature, what does the shift favour?



A) Greater oxygen unloading to tissue
B) Greater oxygen affinity at all pO_2
C) Complete loss of haemoglobin
D) Oxygen release only in the lungs

Q6 One primary spermatocyte completes meiosis normally. How many haploid spermatids are produced, and what is each one's chromosome count in humans?
A) Two, 23
B) Four, 23
C) Four, 46
D) Eight, 23

Q7 The figure shows homologues AB/ab in a double heterozygote. Which two gametes are recombinant after a crossover between loci A and B?



Homologous chromosomes before crossing over

A) The unchanged parental pair AB and ab
B) AB plus Ab
C) Ab and aB
D) The pair ab and aB with parental ab

Q8 A microbial preparation selectively dissolves fibrin clots after myocardial infarction. Which product and source pairing is appropriate?
A) Cyclosporin A from Trichoderma polysporum
B) Statins from Monascus purpureus
C) Penicillin from Penicillium notatum
D) Streptokinase from Streptococcus

Q9 A crop expresses a Bt toxin gene active against a particular insect larva. What determines whether the larva is susceptible after eating the crop?
A) Activation and binding of toxin in the larval gut
B) Whether the crop is a monocot
C) Presence of chlorophyll in the insect
D) The insect’s ability to photosynthesise

Q10 Two forests have equal species richness, but forest X contains many endemic species threatened by habitat loss and forest Y mostly widespread species. Which observation strengthens the case for prioritising X for in-situ conservation?
A) Every species in Y is also endemic
B) Loss of X would remove unique lineages from their native habitat
C) Equal richness guarantees equal extinction risk
D) Ex-situ preservation always maintains the entire ecosystem

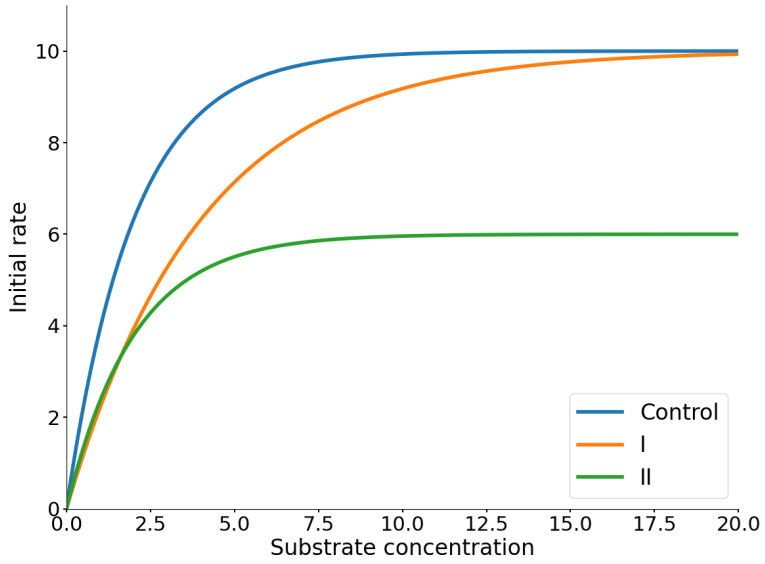
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Q11 A father has an X-linked recessive trait and the mother is homozygous unaffected. Which statement about their daughters is correct?
A) All daughters are affected
B) Half the daughters are affected
C) All daughters inherit the father’s variant X and are carriers
D) No daughter inherits any paternal X

Q12 A freely filtered marker I is neither reabsorbed nor secreted. Its plasma concentration is 0.50 mg mL^{-1} and urine concentration 60 mg mL^{-1}. Urine flow is 1.00 mL min^{-1}. In the same sample, solute G has plasma concentration 2.00 mg mL^{-1} and urine concentration 60 mg mL^{-1}, with no secretion. What are the filtration rate and fraction of filtered G reabsorbed?
A) 20 mL min^{-1}; 3/4
B) 120 mL min^{-1}; 1/4
C) 60 mL min^{-1}; 1/2
D) 120 mL min^{-1}; 3/4

Q13 A neurotransmitter opens ligand-gated Na^+ channels on the postsynaptic membrane. What immediate electrical effect is most likely if Na^+ enters down its electrochemical gradient?
A) Depolarisation of the postsynaptic membrane
B) Hyperpolarisation from K^+ efflux
C) No membrane-potential change because ligand binding consumes ATP
D) Repolarisation from chloride exit

Q14 In the figure, enzyme curve I is shifted to higher substrate concentration at half-maximal speed but reaches the same maximum rate as control. Curve II reaches a lower maximum even at high substrate. Which assignment fits?



A) I and II both competitive
B) I competitive; II noncompetitive
C) I noncompetitive; II competitive
D) Neither shows inhibition

Q15 After ovulation, corpus luteum progesterone keeps the uterine lining secretory. If no implantation occurs, what event most directly precedes menstruation?
A) Continued rise of progesterone
B) Fertilisation in the uterus
C) Degeneration of corpus luteum and fall in progesterone
D) Permanent arrest of endometrial shedding

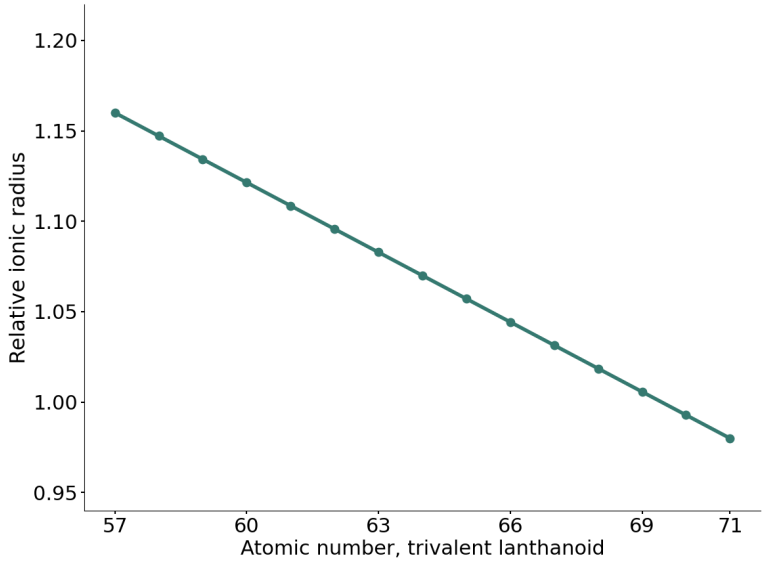
Chemistry

Q16 A neutral molecule XF_4 has six electron domains around its central atom X, of which four are X–F bonds. Assuming equivalent lone pairs and minimal repulsion, what molecular shape and central lone-pair count follow?
A) Square planar, two lone pairs
B) Tetrahedral, two lone pairs
C) Seesaw, one lone pair
D) Square pyramidal, one lone pair

Q17 For the same metal ion, replacing six monodentate ligands by three bidentate ligands can make the complex more stable. Which explanation is most defensible when donor-atom chemistry is comparable?
A) The metal oxidation state must decrease
B) The chelate effect often gives a favourable entropy change from ligand exchange
C) Bidentate ligands have no coordination bonds
D) Stability is controlled solely by the number of metal ions

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Q18 The figure shows a broadly decreasing ionic radius across trivalent lanthanoid ions as atomic number grows. Which cause best fits the trend?



A) Every ion loses an additional proton
B) 4f electrons shield nuclear charge exceptionally well
C) All ions become divalent
D) Added 4f electrons shield poorly, increasing effective nuclear attraction

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Q19 In hydrogen, compare wavelengths of the 3→2 and 2→1 emission lines. What is $\lambda(3\to2)/\lambda(2\to1)$?
A) 5/27
B) 9/4
C) 27/5
D) 3/2

Q20 Among O^{2-}, F^-, Na^+ and Mg^{2+}, all with ten electrons, which order of decreasing ionic radius is expected?
A) $O^{2-}>F^->Na^+>Mg^{2+}$
B) $Mg^{2+}>Na^+>F^->O^{2-}$
C) $F^->O^{2-}>Na^+>Mg^{2+}$
D) $Na^+>Mg^{2+}>O^{2-}>F^-$

Q21 100 mL of 0.40 M Na_2CO_3 is diluted to 400 mL. A 100 mL aliquot is titrated by adding 15 mmol HCl to the well-mixed carbonate solution. Use the ideal two-stage model: carbonate first becomes bicarbonate, and further acid converts bicarbonate to CO_2 and water. Neglect dissolved CO_2 and side equilibria. What mass of CO_2 does this model predict?
A) 0.44 g
B) 0.22 g
C) 0.33 g
D) 1.76 g

Q22 Phenol is more acidic than cyclohexanol under comparable conditions. Which comparison of their conjugate bases best explains this?
A) Cyclohexoxide is resonance stabilised over an aromatic ring
B) Phenoxide cannot carry negative charge on oxygen
C) Phenoxide delocalises negative charge into the ring whereas cyclohexoxide lacks that resonance
D) Phenol releases a hydride ion instead of a proton

Q23 In $CH_3COOH+C_2H_5OH \rightleftharpoons CH_3COOC_2H_5+H_2O$, what change favours more ester at equilibrium without changing temperature?
A) Add water
B) Remove ethanol
C) Add ester
D) Continuously remove water

Q24 Anisole (C_6H_5–O–CH_3) is heated with excess HI. Which main organic products arise from cleavage of its ether bond?
A) Phenol and CH_3I
B) Iodobenzene and methanol
C) Benzene and formaldehyde
D) Cyclohexanol and CH_3I

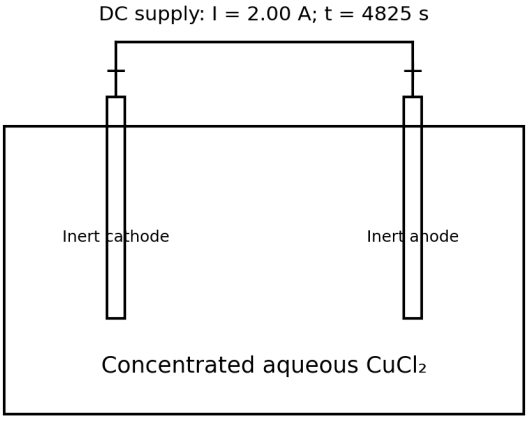
Q25 Direct treatment of aniline with bromine water can give 2,4,6-tribromoaniline. Which sequence best favours mainly the para monobromo derivative after deprotection?
A) Oxidise the ring, then reduce it
B) Acetylate –NH_2, brominate, then hydrolyse the amide
C) Brominate three times, then add more bromine
D) Remove the amino group before electrophilic substitution

Q26 In double-stranded DNA, 30% of all bases are adenine. What fraction are guanine?
A) 10%
B) 30%
C) 20%
D) 40%

Q27 For a liquid with xA=xB=0.5, pA°=100 kPa and pB°=50 kPa, total vapour pressure is measured as 60 kPa. Relative to ideal Raoult behaviour, what deviation and interaction pattern does this support?
A) +15 kPa; weaker unlike attractions
B) −15 kPa; weaker unlike attractions
C) +15 kPa; stronger unlike attractions
D) −15 kPa; stronger unlike attractions

Q28 The diagram shows an electrolytic cell containing concentrated aqueous copper(II) chloride with inert electrodes. It operates at 2.00 A for 4825 s. Assume 100% current efficiency, copper deposition at the cathode and chloride discharge at the anode; $F = 96500$ C mol^{-1}, $M(\mathrm{Cu}) = 63.5$ g mol^{-1}. What are the copper mass deposited and chlorine amount evolved?

Electrolysis: measured current and time



A) 3.175 g and 0.050 mol
B) 6.35 g and 0.100 mol
C) 3.175 g and 0.100 mol
D) 6.35 g and 0.050 mol

Q29 Two Arrhenius reactions have the same pre-exponential factor at one fixed temperature T. Reaction X has activation energy E, while Y has E+RT ln 2. What is kY/kX?
A) 2
B) 1/2
C) 1
D) 1/4

Q30 For A(g)+B(g) ⇌ C(g), Kc=2 M^{-1}. At an instant [A]=[B]=1 M and [C]=4 M. In which direction will the reaction initially shift?
A) Forward because reactants are equal
B) No shift because total concentration is 6 M
C) Forward because Qc exceeds Kc
D) Reverse because Qc=4 M^{-1} exceeds Kc

Mathematics

Q31 For what value of c does $x^2+y^2-4x+6y+c=0$ touch the x-axis at exactly one point?
A) 4
B) 9
C) 13
D) -4

Q32 On which open intervals is $f(x)=x^3-3x$ strictly increasing?
A) $(-1,1)$ only
B) $(-\infty,-1)$ and $(1,\infty)$
C) $(-\infty,1)$ only
D) All real numbers except ±1

Q33 For $a>0$, $\int_0^a 2x/(1+x^2)^2\,dx=3/4$. What is a?
A) 1
B) 2
C) $\sqrt{3}$
D) $\sqrt{2}$

Q34 For $f(t)=\sqrt{(t-1)}$ and $g(x)=1/(3-x)$, what is the real domain of $f(g(x))$?
A) $x<3$
B) $x\geq2$
C) $x\leq2$
D) $[2,3)$

Q35 A 3×3 matrix A has determinant 5. Matrix B is formed by replacing row 2 by row 2+3(row 1). What is det B?
A) 5
B) 15
C) 20
D) 0

Q36 Box A is chosen with probability 2/3 and holds 2 red, 1 blue balls. Box B is chosen with probability 1/3 and holds 3 red, 1 blue balls. Two balls are drawn without replacement from the chosen box and both are red. What is the posterior probability of box B?
A) 1/3
B) 3/7
C) 1/2
D) 4/7

Q37 How many distinct arrangements of the letters of MISSISSIPPI place all four I's consecutively?
A) 1680
B) 420
C) 3360
D) 840

Q38 If $C(n,3)=C(n,4)$ for integer $n\geq4$, what is n?
A) 6
B) 8
C) 7
D) 9

Q39 If the sum of the first n terms is $S_n=3n^2+2n$ for $n\geq1$ and $S_0=0$, what is the fifth term?
A) 29
B) 25
C) 35
D) 31

Q40 A real a makes L=lim as x→0 of [e^(2x)−1−2x−a x²]/[e^x−1−x−x²/2] finite. What is the pair (a,L)?
A) (4,4)
B) (2,4)
C) (4,8)
D) (2,8)

Q41 The family of curves $y^2=4a(x+b)$, with nonzero a and arbitrary b, has which differential equation after eliminating both constants?
A) $yy''-(y')^2=0$
B) $yy''+(y')^2=0$
C) $y''=0$
D) $yy'=0$

Q42 A line has direction vector (1,2,2). What is the sine of its acute angle with the xy-plane?
A) 1/3
B) √5/3
C) 2/3
D) 1

Q43 Points A(1,2,3), B(a,4,5) and C(5,6,7) are collinear with B between A and C. What are (a, AB:BC)?
A) (2, 1:1)
B) (3, 1:2)
C) (4, 1:1)
D) (3, 1:1)

Q44 Values 0,2,4 occur with frequencies 1,2,1. What is mean deviation about the mean?
A) 1
B) 2
C) 1/2
D) 4/3

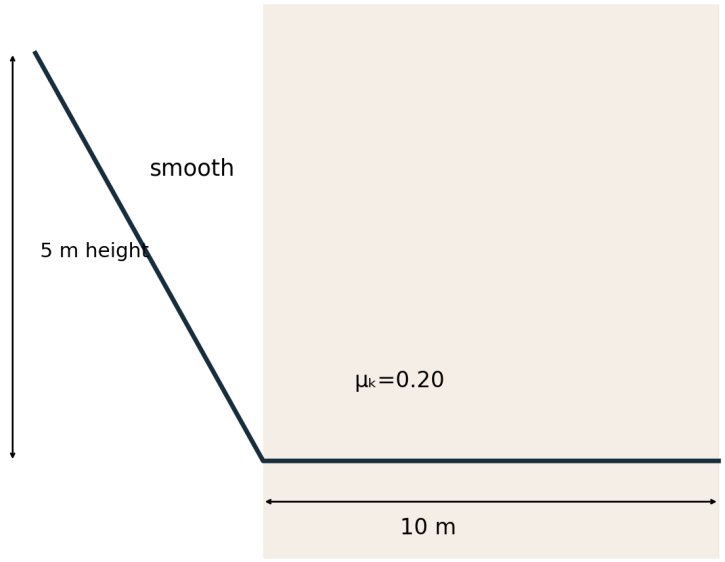
Q45 How many distinct x in [0,4π] satisfy sin x=cos(2x)?
A) 4
B) 5
C) 6
D) 8

Physics

Q46 A particle starts at rest, accelerates at +2 m/s² for 2 s, then at −1 m/s² for 4 s. What displacement occurs in the full 6 s?
A) 12 m
B) 8 m
C) 4 m
D) 16 m

Q47 A 60 kg person on a scale in an elevator reads 720 N while accelerating upward. With g=10 m/s², what will the scale read if acceleration reverses with the same magnitude?
A) 600 N
B) 480 N
C) 720 N
D) 360 N

Q48 As shown, a block starts from rest 5 m above a frictionless track, then traverses 10 m of horizontal rough track with μk=0.20. For g=10 m/s², what is v² at the rough segment's end?



A) 100 m²/s²
B) 80 m²/s²
C) 60 m²/s²
D) 20 m²/s²

Q49 A rigid rotor's applied torque and moment of inertia are each doubled. What happens to its angular acceleration, assuming both torques act in the same rotational sense?
A) It quadruples
B) It doubles
C) It halves
D) It remains unchanged

Q50 For a uniform-density spherical planet of radius R, what is g at r=R/2 relative to surface value g(R)?
A) g(R)/2
B) g(R)/4
C) 2g(R)
D) 4g(R)

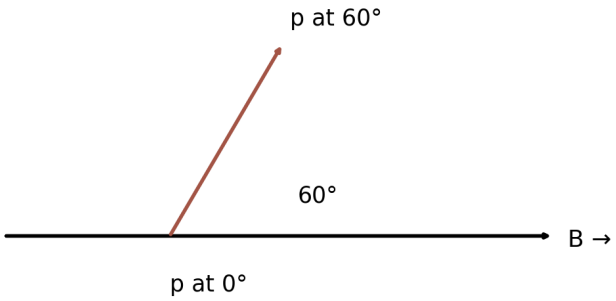
Q51 Two spheres of the same material settle slowly in one viscous liquid under Stokes drag. If one radius is twice the other, what is the ratio of their terminal speeds (larger/smaller)?
A) 2
B) 4
C) 8
D) 16

Q52 An ideal gas with γ=5/3 expands reversibly and adiabatically to twice its initial volume. What is Tf/Ti?
A) 1/2
B) 2^(2/3)
C) 2^(−2/3)
D) 2^(−5/3)

Q53 A closed Gaussian surface encloses +q, −q and +2q. What is its net outward electric flux?
A) Zero
B) q/ε_0
C) $4q/\varepsilon_0$
D) $2q/\varepsilon_0$

Q54 Capacitors C and 2C are connected in series to an ideal battery of voltage V. What is the magnitude of charge on the 2C capacitor?
A) 2CV/3
B) CV/3
C) CV
D) 2CV

Q55 The figure shows a magnetic dipole p rotating slowly from aligned (0°) to 60° relative to uniform B. What work must an external agent do quasistatically?

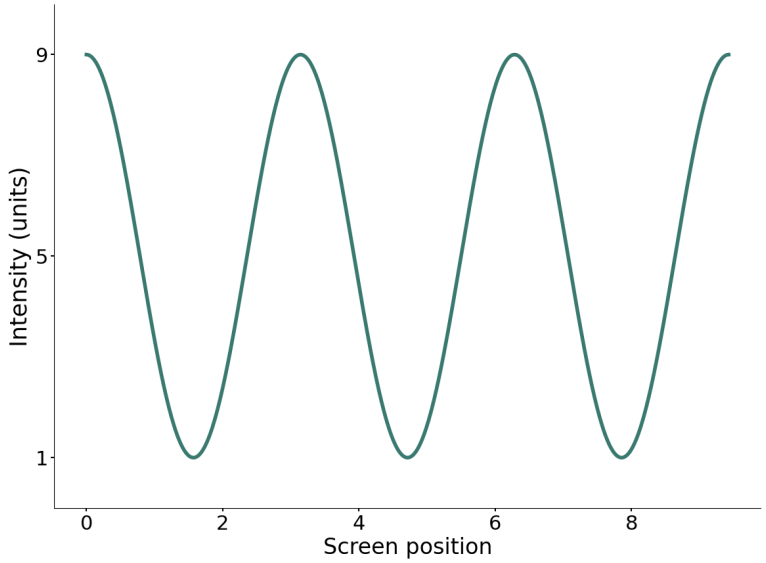


A) $-pB/2$
B) $+pB/2$
C) $+pB$
D) Zero

Q56 For two coils with mutual inductance 0.50 H, primary current grows at +2 A/s in the chosen positive direction. What is induced secondary emf under the matching positive flux convention?
A) +1 V
B) −2 V
C) −1 V
D) +2 V

Q57 A thin convex lens of focal length 10 cm has an object 30 cm to its left. A plane mirror is placed d cm to its right, before the first-pass image at 15 cm. For the reflected rays emerging back through the lens to be parallel, what is d?
A) 5 cm
B) 10 cm
C) 15 cm
D) 12.5 cm

Q58 The figure shows two-beam interference with Imax=9 units and Imin=1 unit. What is the fringe visibility?



A) 1/2
B) 1
C) 1/5
D) 4/5

Q59 For a hydrogen emission line terminating at n=2, its wavenumber is 3R/16 (R is Rydberg constant). From which initial level did the electron fall?
A) n=3
B) n=4
C) n=5
D) n=6

Q60 Two pure radioisotopes initially have equal activities. Their half-lives are 2 h and 4 h, respectively. What is the ratio of first to second activities after 4 h?
A) 1:4
B) 2:1
C) 1:2
D) 1:1

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A Angiosperm seed character

Answer: (A) Monocot angiosperm.

Why: An enclosed ovule identifies angiosperms; one cotyledon and parallel venation independently support the monocot group. Both sets of evidence must agree. The enclosed ovule identifies the flowering-plant lineage before any vegetative clue is used. The single embryonic cotyledon and parallel venation independently point to the monocot side of that lineage.

Why not the others: (B) and (C) contradict ovary enclosure; (D) contradicts cotyledon number and typical venation.

Key point: Use floral origin and embryo characters together.

Q2 / B Pulmonary versus systemic venous return in a frog

Answer: (B) Lung return to the left atrium falls; systemic return through the sinus venosus is not directly blocked.

Why: The pulmonary veins connect the lungs to the left atrium, so the obstruction acts on that route first. Cutaneous return is carried into the systemic venous route toward the sinus venosus and right atrium, rather than being assigned to pulmonary veins. Keeping systemic veins open does not guarantee unchanged later cardiac output, but it distinguishes the directly obstructed pathway.

Why not the others: (A) assigns skin return to the wrong vessel; (C) reverses the atrial receiving pathways; (D) incorrectly extends a pulmonary-vein obstruction to every venous entrance.

Key point: Trace each vessel to its receiving chamber before predicting consequences.

Q3 / C Organelle fractionation marker enzymes

Answer: (C) Peroxisome.

Why: Catalase decomposes hydrogen peroxide generated by peroxisomal oxidases. Low chlorophyll and cytochrome oxidase help exclude chloroplast and mitochondrial contamination as the principal source. Catalase detoxifies peroxide inside peroxisomes, making it a functional marker rather than a generic protein. Low chlorophyll and respiratory oxidase are independent negative controls against chloroplast or mitochondrial enrichment.

Why not the others: (A) is marked by chlorophyll; (B) by respiratory-chain oxidases; (D) contains chromosomes rather than this characteristic catalase activity.

Key point: Identify an organelle by a marker plus negative controls.

Q4 / D Stomatal closure and gas exchange

Answer: (D) CO_2 from air into intercellular spaces.

Why: Stomata are the principal gas-exchange pores; closing them raises diffusion resistance for external CO_2, limiting substrate supply to photosynthetic cells. Other processes can respond secondarily. The barrier is at the leaf surface where gas exchange with air occurs. Reduced CO_2 supply can subsequently reduce carboxylation, but the question asks for the *immediate entry flux*, not a downstream metabolic rate.

Why not the others: (A) occurs in roots; (B) is phloem loading; (C) is not the uniquely direct stomatal entry flux.

Key point: Locate the anatomical barrier before inferring metabolic consequences.

Q5 / A Haemoglobin oxygen dissociation shift

Answer: (A) Greater oxygen unloading to tissue.

Why: Higher CO_2 and temperature shift the dissociation relation rightward, lowering saturation at the same tissue oxygen pressure. More bound oxygen can therefore be released during passage through active tissue. Compare the two curves at the dashed common tissue pO_2, not at different x positions. Curve B leaves less oxygen bound after tissue passage; the saturation difference is oxygen available for unloading.

Why not the others: (B) reverses the affinity shift; (C) is unsupported by a curve shift; (D) ignores tissue delivery.

Key point: Compare saturation at the same pO_2, not just the apparent direction of a graph.

Q6 / B Human gametogenesis chromosome number

Answer: (B) Four, 23.

Given: A human primary spermatocyte is diploid with 46 chromosomes.

Formula/Principle: Two successive meiotic divisions give four haploid products from one primary cell.

Step 1: Meiosis I separates homologues and meiosis II separates sister chromatids.

Step 2: Four spermatids result, each with n=23.

Why: Chromosome number halves while the number of descendant cells quadruples. Meiosis I produces two haploid secondary spermatocytes, each still carrying duplicated chromosomes. Meiosis II separates sister chromatids to give four spermatids with $n = 23$, while fertilisation would restore diploidy later.

Why not the others: (A) stops after meiosis I; (C) retains diploidy; (D) adds an extra division.

Key point: Track cells and chromosomes separately.

Q7 / C Linkage phase and recombinant gametes

Answer: (C) Ab and aB.

Why: The starting coupling-phase parental combinations are AB and ab. Exchanging chromosome segments between the loci creates the mixed combinations Ab and aB. The top chromatid's AB and bottom chromatid's ab are the original combinations. A crossover between the marked loci swaps distal segments, producing Ab and aB; no crossover outside the interval would change this phase.

Why not the others: (A) are both parental; (B) and (D) mix one parental and one recombinant type.

Key point: Establish parental phase before labelling products of crossing over.

Q8 / D Microbial product and application

Answer: (D) Streptokinase from Streptococcus.

Why: Streptokinase activates plasminogen, promoting fibrin breakdown in clots; it is obtained from streptococcal cultures. The requested function is thrombolysis, not immunosuppression or lipid reduction. Streptokinase does not directly become fibrin; it activates the host plasminogen-to-plasmin system, and plasmin digests the clot. That functional chain distinguishes it from antimicrobial and cholesterol-lowering microbial products.

Why not the others: (A) is immunosuppressive; (B) inhibits cholesterol synthesis; (C) is antibacterial.

Key point: Match mechanism and microbial source to the clinical use.

Q9 / A Bt crop tissue specificity

Answer: (A) Activation and binding of toxin in the larval gut.

Why: Bt protoxin becomes active in the insect gut and binds susceptible gut-cell receptors, leading to membrane damage. Expression by the plant alone does not guarantee that every insect species has the required gut conditions and receptors. In a susceptible alkaline larval gut, the protoxin is activated and can bind specific epithelial receptors. Species lacking compatible receptors need not be harmed merely because the plant expresses the same gene.

Why not the others: (B) concerns plant taxonomy, not larval sensitivity; (C) and (D) assign photosynthetic traits to insects.

Key point: Target specificity depends on the insect's response to the toxin.

Q10 / B Conservation evidence and endemism

Answer: (B) Loss of X would remove unique lineages from their native habitat.

Why: Endemic species have restricted natural distributions, so destruction of their habitat can cause global loss rather than a local population decline alone. In-situ protection also preserves ecological interactions. Equal richness counts species but does not capture their geographic restriction. If an endemic population disappears from its only habitat, rescue by an unaffected population elsewhere may be impossible.

Why not the others: (A) contradicts the given contrast; (C) confuses count with distribution and threat; (D) overstates ex-situ coverage.

Key point: Richness and irreplaceability are different conservation criteria.

Q11 / C X-linked recessive transmission

Answer: (C) All daughters inherit the father's variant X and are carriers.

Why: A father passes his only X chromosome to every daughter, while the homozygous unaffected mother supplies a normal X. For a fully recessive trait each daughter is heterozygous. All daughters receive the affected father's X rather than his Y. The unaffected homozygous mother's X supplies a functional allele, masking a recessive phenotype while leaving the daughter heterozygous.

Why not the others: (A) and (B) require affected or carrier maternal X; (D) confuses daughters with sons, who receive paternal Y.

Key point: Track the sex chromosome contributed by each parent.

Q12 / D Filtration marker and separate-solute reabsorption

Answer: (D) 120 $\mathrm{mL\,min^{-1}}$ and $3/4$.

Given: I is filtered without tubular transport, so its filtered load equals its excretion rate.

Formula/Principle:

$$\mathrm{GFR} = \frac{U_I \dot{V}}{P_I}, \quad L_G = \mathrm{GFR}\, P_G, \quad R_G = L_G - U_G \dot{V}.$$

Step 1:

$$\mathrm{GFR} = \frac{60(1.00)}{0.50} = 120\ \mathrm{mL\,min^{-1}}.$$

Step 2:

$$L_G = 120(2.00) = 240\ \mathrm{mg\,min^{-1}}, \quad E_G = 60(1.00) = 60\ \mathrm{mg\,min^{-1}}.$$

Step 3:

$$R_G = 240 - 60 = 180\ \mathrm{mg\,min^{-1}}, \quad R_G/L_G = 180/240 = 3/4.$$

Why: The marker's no-transport property identifies filtration independently of G's handling. With secretion excluded for G, the difference between filtered and excreted mass is net reabsorption, not an unexplained loss of water volume.

Why not the others: (A) miscalculates marker clearance; (B) reports the excreted rather than reabsorbed fraction; (C) confuses concentration with filtration rate and then uses an incorrect load.

Key point: Use a filtration marker first, then balance the second solute's tubular mass flow.

Q13 / A Neural synapse and receptor response

Answer: (A) Depolarisation of the postsynaptic membrane.

Why: Positive Na^+ entry makes the cell interior less negative, creating an excitatory postsynaptic potential. Reaching an action-potential threshold still depends on size and summation of responses. Opening cation channels makes the inside potential move toward the positive sodium equilibrium potential. An excitatory potential may still be subthreshold, so depolarisation does not guarantee an action potential.

Why not the others: (B) and (D) invoke different ion fluxes; (C) ignores the charge movement across the membrane.

Key point: Predict voltage direction from the sign and direction of ion movement.

Q14 / B Competitive versus noncompetitive inhibition

Answer: (B) I competitive; II noncompetitive.

Why: Excess substrate can overcome competition at an active site, retaining Vmax but increasing apparent Km. A pure noncompetitive inhibitor reduces active catalytic capacity, lowering Vmax despite abundant substrate. The control and I curves approach the same upper plateau while I needs more substrate for a given rate, consistent with reversible competition. Curve II's lower plateau cannot be rescued by high substrate under the stated simple model.

Why not the others: (A) cannot explain II's lower plateau; (C) reverses the characteristic patterns; (D) ignores both altered curves.

Key point: Compare saturation plateaus as well as midpoint concentrations.

Q15 / C Menstrual hormone sequence

Answer: (C) Degeneration of corpus luteum and fall in progesterone.

Why: Without embryonic maintenance signals the corpus luteum regresses, withdrawing hormonal support from the endometrium. Spiral vessels and tissue then break down and menstruation follows. Absent implantation, there is no sustained signal maintaining the corpus luteum. Falling progesterone removes support for the secretory endometrium, leading to vascular breakdown and shedding in sequence.

Why not the others: (A) maintains the lining; (B) is not a prerequisite and fertilisation normally occurs in the oviduct; (D) contradicts the stated no-implantation cycle.

Key point: Link luteal regression to progesterone withdrawal and lining breakdown.

Chemistry

Q16 / A VSEPR and lone-pair count

Answer: (A) Square planar, two lone pairs.

Why: Six electron domains form an octahedral arrangement. Four bonds leave two lone pairs, and placing them opposite each other minimises lone-pair repulsion; the remaining four bonds lie in one square plane. An octahedral electron-domain arrangement has six positions. Placing the two lone pairs trans to each other maximises their separation and leaves the four fluorines in a square, not a tetrahedron.

Why not the others: (B) confuses molecular with electron geometry; (C) has five domains; (D) also requires five domains.

Key point: Count all electron domains before removing lone-pair positions from the molecular

shape.

Q17 / B Chelate stability and ligand replacement

Answer: (B) The chelate effect often gives a favourable entropy change from ligand exchange.

Why: Multidentate ligands form rings with several attachments to the same metal; exchange can liberate more free particles into solution and resist complete dissociation. This comparison assumes similar donor chemistry, so it does not prove every chelate always wins. Chelating ligands bind through two donors to the same metal and form rings. Their dissociation requires breaking coordinated contacts, while exchange can increase the number of free particles and thus favour complex formation entropically.

Why not the others: (A) is not required; (C) contradicts coordination; (D) ignores ligand binding.

Key point: Compare exchange equilibria, not just bond counts.

Q18 / D Lanthanoid contraction

Answer: (D) Added 4f electrons shield poorly, increasing effective nuclear attraction.

Why: As protons are added, electrons enter 4f subshells that do not fully screen the increasing nuclear charge from outer electrons, drawing the ionic shell inward. The comparison holds oxidation state fixed at +3. As atomic number grows, proton count increases while the newly added 4f electrons provide relatively weak shielding. The same +3 charge across the graph makes the radius decline meaningful rather than a change in oxidation state.

Why not the others: (A) reverses nuclear change; (B) predicts weak contraction; (C) violates the trivalent label.

Key point: Compare like-charge ions while tracking shielding effectiveness.

Q19 / C Hydrogen transition wavelength ratio

Answer: (C) 27/5.

Given: The Rydberg wavenumbers are R(1/4−1/9)=5R/36 and R(1−1/4)=3R/4.

Formula/Principle: Wavelength is inverse to wavenumber.

Step 1: Ratio λ32/λ21=(3/4)/(5/36).

Step 2: Simplify to 27/5.

Worked derivation:

$$\widetilde{\nu}_{32} = R(1/4 - 1/9) = 5R/36,$$
$$\widetilde{\nu}_{21} = R(1 - 1/4) = 3R/4,$$
$$\lambda_{32}/\lambda_{21} = \widetilde{\nu}_{21}/\widetilde{\nu}_{32} = (3/4)/(5/36) = 27/5.$$

Why: The lower-energy Balmer photon has the longer wavelength. Hydrogen's wavenumber is the difference of inverse-square level terms, not the difference of the level labels. Since $\tilde{\nu}_{32} < \tilde{\nu}_{21}$, the 3→2 wavelength must be longer, checking the ratio's direction.

Why not the others: (A) is the inverse; (B) and (D) confuse level-number ratios with spectral differences.

Key point: Compute energy differences before comparing wavelengths.

Q20 / A Isoelectronic ionic radii

Answer: (A) O^{2-}>F^{-}>Na^{+}>Mg^{2+}.

Why: All have the same electron count and broadly similar electronic arrangement. Nuclear charge increases from O to Mg, pulling that electron cloud closer and shrinking the ion. All four ions have neon-like ten-electron configurations, so electron count is controlled. Increasing nuclear charge pulls essentially the same electron cloud inward, ordering O^{2-} largest through Mg^{2+} smallest.

Why not the others: (B) reverses the trend; (C) swaps O and F; (D) incorrectly places the cations above the anions.

Key point: Hold electron number constant when comparing effective nuclear pull.

Q21 / B Aliquot and staged carbonate neutralisation

Answer: (B) 0.22 g.

Given: Original carbonate $= 0.100(0.40) = 0.040$ mol; the aliquot contains one quarter of it, or 10 mmol.

Formula/Principle:

$$CO_3^{2-} + H^+ \longrightarrow HCO_3^-, \qquad HCO_3^- + H^+ \longrightarrow CO_2 + H_2O.$$

Step 1: The first 10 mmol acid converts all 10 mmol carbonate to bicarbonate.

Step 2: Acid remaining $= 15 - 10 = 5$ mmol; it produces 5 mmol CO_2, leaving 5 mmol bicarbonate.

Step 3:

$$m(CO_2) = 0.005(44) = 0.22 \text{ g}.$$

Why: The first proton does not itself release one molecule of carbon dioxide in the stipulated sequential model. Applying the overall two-proton equation to the acid alone while carbonate remains available misses the bicarbonate intermediate and gives the wrong gas amount.

Why not the others: (A) converts the whole aliquot to gas despite insufficient acid; (C) uses half of all added acid as gas without accounting for the first stage; (D) uses carbonate in the whole stock rather than the aliquot.

Key point: Track aliquot moles and successive protonation stages separately.

Q22 / C Resonance and acidity

Answer: (C) Phenoxide delocalises negative charge into the ring whereas cyclohexoxide lacks that resonance.

Why: Acid strength depends on the stability of the conjugate base. Phenoxide has resonance contributors involving the aromatic π system; the saturated cyclohexyl ring does not provide analogous delocalisation. Deprotonation tests the stability of the *products* of acid dissociation. Phenoxide can distribute negative charge into an aromatic π system, whereas cyclohexoxide remains localised on oxygen in a saturated ring.

Why not the others: (A) assigns aromatic resonance to a saturated ring; (B) denies the principal oxygen-centred structure; (D) describes the wrong leaving species.

Key point: Compare conjugate-base stabilisation rather than only neutral structures.

Q23 / D Esterification equilibrium

Answer: (D) Continuously remove water.

Why: Water is a product; removing it lowers the reaction quotient and the system responds by forming more ester and water. This changes equilibrium composition without asserting a change in equilibrium constant at fixed temperature. With K fixed at unchanged temperature, removing water lowers the product-side reaction quotient. The system then proceeds forward until a new equilibrium composition restores Q=K; the equilibrium constant itself is unchanged.

Why not the others: (A) and (C) add products; (B) removes a reactant, each favouring the reverse direction.

Key point: Distinguish a composition shift from changing K.

Q24 / A Aryl-alkyl ether cleavage

Answer: (A) Phenol and CH_3I.

Why: After ether oxygen is protonated, iodide attacks the accessible methyl carbon by SN2, breaking the O–CH_3 bond. The aryl C–O bond has partial double-bond character and is not cleaved by ordinary backside attack at sp^2 carbon. Iodide can attack the unhindered methyl carbon after protonation of ether oxygen. Direct SN2 attack at the sp^2 aromatic carbon cannot follow the same trajectory, so the phenolic C–O link remains.

Why not the others: (B) assumes aryl substitution; (C) and (D) change the aromatic skeleton.

Key point: In aryl–alkyl ethers the alkyl side is normally cleaved.

Q25 / B Aniline bromination and protection

Answer: (B) Acetylate –NH_2, brominate, then hydrolyse the amide.

Why: Acyl protection reduces the amino group's strong activation while retaining ortho/para direction; steric effects favour para substitution. Hydrolysis restores –NH_2 after controlled substitution. The acetyl group decreases lone-pair donation into the ring without eliminating directing influence. Once mainly para bromination occurs, hydrolysis restores the amino group while keeping the installed bromine position.

Why not the others: (A) does not selectively moderate activation; (C) worsens multiple substitution; (D) loses the directing group needed for the target.

Key point: Temporarily reduce activation to control substitution count.

Q26 / C Double-stranded DNA base pairing

Answer: (C) 20%.

Given: A=30% across both strands.

Formula/Principle: Chargaff pairing gives A=T and G=C for double-stranded DNA.

Step 1: T=30%, leaving 40% for G+C.

Step 2: G=C=20%.

Worked derivation:

$$A = T = 0.30, \quad G + C = 1 - 0.60 = 0.40, \quad G = C = 0.20.$$

Why: Complementary pairing applies to the combined double-stranded composition. Equal amounts of A and T occupy 60% of all nucleotides, leaving 40% jointly for G and C. Equality of G and C divides that remainder in half; the inference requires a double-stranded paired sample.

Why not the others: (A) halves twice; (B) assigns all bases the A share; (D) gives all the residual share to G.

Key point: Account for both members of each base pair.

Q27 / D Deviation from ideality and interactions

Answer: (D) −15 kPa; stronger unlike attractions.

Why: The ideal total is 0.5×100+0.5×50=75 kPa. Measured minus ideal is 60−75=−15 kPa; a lower escape tendency is consistent with stronger A–B attractions than like-particle interactions. The Raoult benchmark is $0.5(100)+0.5(50)=75$ kPa; measured 60 kPa is 15 kPa lower. Stronger unlike attractions are a plausible molecular cause, not a proof of a unique bonding structure.

Why not the others: (A) and (C) reverse the measured sign; (B) assigns the opposite interaction trend, which usually raises vapour pressure.

Key point: Calculate the ideal benchmark before diagnosing an interaction mechanism.

Q28 / A Faraday charge balance at two electrodes

Answer: (A) 3.175 g copper and 0.050 mol chlorine.

Given: $I = 2.00$ A, $t = 4825$ s; the electrode products and efficiency are explicitly stipulated.

Formula/Principle:

$$\mathrm{Cu^{2+}} + 2\,\mathrm{e^-} \longrightarrow \mathrm{Cu}, \qquad 2\,\mathrm{Cl^-} \longrightarrow \mathrm{Cl_2} + 2\,\mathrm{e^-}.$$

Step 1:

$$Q = It = 2.00(4825) = 9650\ \mathrm{C}.$$

Step 2:

$$n_e = Q/F = 0.100\ \mathrm{mol}.$$

Step 3: Each product needs two moles of electrons per mole, hence $n_{\mathrm{Cu}} = n_{\mathrm{Cl_2}} = 0.0500$ mol.

Step 4:

$$m_{\mathrm{Cu}} = 0.0500(63.5) = 3.175\ \mathrm{g}.$$

Why: Equal charge passes both electrodes, but electron stoichiometry converts that charge to chemical amounts. The specified products remove ambiguity from concentration-dependent competition with water; the diagram supplies actual apparatus and operating quantities rather than invented discharge-preference bars.

Why not the others: (B) omits both two-electron factors; (C) omits only the chlorine factor; (D) omits only the copper factor.

Key point: Convert charge to electron moles before applying each half-reaction.

Q29 / B Rate response to activation energy

Answer: (B) 1/2.

Given: Equal pre-exponential factors and activation-energy difference RT ln2.

Formula/Principle: k=A exp(−Eₐ/RT).

Step 1: Divide the two rate constants: kY/kX=exp[−(RT ln2)/(RT)].

Step 2: exp(−ln2)=1/2.

Worked derivation:

$$\frac{k_Y}{k_X} = \frac{Ae^{-(E_X + RT\ln 2)/(RT)}}{Ae^{-E_X/(RT)}} = e^{-\ln 2} = \frac{1}{2}.$$

Why: An extra barrier suppresses rate at the same temperature; the unknown common A cancels. The Arrhenius exponent changes by $-\Delta E_a/(RT) = -\ln 2$. Equal prefactors cancel, leaving $e^{-\ln 2} = 1/2$; the sign also makes physical sense because a higher barrier slows the reaction.

Why not the others: (A) reverses the sign; (C) ignores barrier change; (D) doubles the exponent.

Key point: Compare Arrhenius rates as a ratio.

Q30 / D Reaction quotient after composition change

Answer: (D) Reverse because Qc=4 M^{-1} exceeds Kc.

Given: Current concentrations A=B=1 M and C=4 M.

Formula/Principle: Qc=[C]/([A][B]); if Q>K, reverse reaction lowers Q.

Step 1: Qc=4/(1×1)=4 M^{-1}.

Step 2: Since 4>2, C is consumed initially.

Worked derivation:

$$Q_c = \frac{[\mathrm{C}]}{[\mathrm{A}][\mathrm{B}]} = \frac{4}{1(1)} = 4\ \mathrm{M}^{-1} > K_c = 2\ \mathrm{M}^{-1}.$$

Why: The instantaneous composition has too much product relative to equilibrium. At the stated instant $Q_c = 4\,\mathrm{M}^{-1}$ while $K_c = 2\,\mathrm{M}^{-1}$. Consuming C and generating A and B lowers the product-to-reactant ratio toward K; equal initial reactant concentrations do not imply equilibrium.

Why not the others: (A) and (B) ignore the mass-action expression; (C) reverses the Q>K implication.

Key point: Evaluate Q with the current composition, then compare with K.

Mathematics

Q31 / A Tangency of parameterised circle to axis

Answer: (A) 4.

Given: Completing squares yields $(x-2)^2+(y+3)^2=13-c$, centre $(2,-3)$.

Formula/Principle: Tangency to x-axis requires radius equal to centre-to-axis distance 3.

Step 1: $r^2=9$.

Step 2: $13-c=9$, hence $c=4$.

Worked derivation:

$$(x-2)^2+(y+3)^2 = 13-c, \quad r = |-3| = 3, \quad 13-c = r^2 = 9 \Longrightarrow c = 4.$$

Why: A real nonzero radius is also confirmed. Completing the square identifies the centre before comparing it with the x-axis. The resulting radius is real and nonzero, so the circle has a single contact point.

Why not the others: (B) gives radius 2; (C) zero radius; (D) a larger circle crossing the axis twice.

Key point: Translate the geometric tangency into a radius condition.

Q32 / B Monotonicity from derivative sign

Answer: (B) $(-\infty,-1)$ and $(1,\infty)$.

Given: $f'(x)=3x^2-3$.

Formula/Principle: Positive derivative implies local strict increase on intervals.

Step 1: $f'>0$ when $x^2>1$.

Step 2: This gives $x<-1$ or $x>1$; between them $f'<0$.

Worked derivation:

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1) > 0 \Longleftrightarrow x < -1 \text{ or } x > 1.$$

Why: Stationary points separate three sign regions, not a single punctured increasing domain. The derivative is negative on the entire middle interval, not merely at a point. The isolated zeros mark changes in monotonicity, so solve the sign inequality over intervals.

Why not the others: (A) is decreasing; (C) includes it; (D) ignores negative derivative throughout $(-1,1)$.

Key point: Make a sign chart rather than checking only derivative zeros.

Q33 / C Recover an upper bound from a definite integral

Answer: (C) √3.

Given: u=1+x² transforms the lower and upper bounds to 1 and 1+a².

Formula/Principle: ∫u⁻²du=−1/u.

Step 1: Integral=1−1/(1+a²). Set it equal to 3/4.

Step 2: 1/(1+a²)=1/4, giving a²=3; a>0 selects √3.

Worked derivation:

$$u = 1 + x^2, \quad \int_0^a \frac{2x\,dx}{(1+x^2)^2} = \int_1^{1+a^2} u^{-2}\,du = 1 - \frac{1}{1+a^2} = \frac{3}{4} \Longrightarrow a = \sqrt{3}.$$

Why: The measured area fixes the bound rather than asking for a routine value at supplied endpoints. The upper bound changes under substitution and remains symbolic. The positive-a condition discards the negative square root after solving the resulting equation.

Why not the others: (A), (B), (D) give 1/2, 4/5 and 2/3 respectively.

Key point: Keep the transformed upper limit symbolic until the area condition is applied.

Q34 / D Domain of composition with radical and pole

Answer: (D) [2,3).

Given: g must be defined and g(x)≥1 for the square root.

Formula/Principle: Solve 1/(3−x)≥1 with x≠3 by sign intervals.

Step 1: For x>3 the quotient is negative and cannot reach 1.

Step 2: For x<3, multiply by positive 3−x to obtain 1≥3−x, so x≥2.

Worked derivation:

$$g(x) \geq 1,\ x \neq 3; \quad \frac{1}{3-x} \geq 1, \quad x < 3 \Rightarrow x \geq 2; \qquad \mathrm{dom}(f \circ g) = [2, 3).$$

Why: x=2 gives radical zero and is included; x=3 is a pole and excluded. The inner rational expression must exist and then satisfy the outer square-root threshold. Multiplying an inequality without first fixing the denominator's sign could reverse it.

Why not the others: (A) admits x<2; (B) includes the pole and beyond; (C) reverses the bound.

Key point: Impose inner-function definition before outer-function domain.

Q35 / A Determinant under row replacement

Answer: (A) 5.

Given: One row is augmented by a multiple of another row.

Formula/Principle: Such a row replacement leaves determinant unchanged.

Step 1: Apply $R_2 \to R_2 + 3R_1$.

Step 2: Therefore det B=det A=5.

Worked derivation:

$$\det(B) = \det(R_1, R_2 + 3R_1, R_3) = \det A + 3\det(R_1, R_1, R_3) = 5 + 0 = 5.$$

Why: Multilinearity adds a term with two equal rows, whose determinant is zero. The extra determinant vanishes because it has two identical rows. This explains the invariant rather than merely recalling a row-operation rule.

Why not the others: (B) and (C) incorrectly scale determinant; (D) assumes dependent rows were created.

Key point: Distinguish row replacement from row scaling.

Q36 / B Bayes posterior from two draws without replacement

Answer: (B) 3/7.

Given: Priors are P(A)=2/3 and P(B)=1/3; the two-red event has different without-replacement likelihoods.

Formula/Principle: Bayes posterior is P(B)P(RR|B) divided by the sum of both prior-weighted likelihoods.

Step 1: P(RR|A)=(2/3)(1/2)=1/3; P(RR|B)=(3/4)(2/3)=1/2.

Step 2: B's weighted likelihood is 1/6, A's is 2/9; posterior (1/6)/(1/6+2/9)=3/7.

Worked derivation:

$$P(RR \mid A) = \frac{2}{3}\frac{1}{2} = \frac{1}{3}, \quad P(RR \mid B) = \frac{3}{4}\frac{2}{3} = \frac{1}{2}; \quad P(B \mid RR) = \frac{(1/3)(1/2)}{(2/3)(1/3) + (1/3)(1/2)} = \frac{3}{7}.$$

Why: The second draw depends on the first, so multiplying a single-draw red probability by itself would be wrong. Without replacement changes the second-draw probability in each box. The posterior compares prior-weighted two-draw likelihoods, not the raw red counts.

Why not the others: (A) is only the prior; (C) ignores the unequal priors; (D) reverses posterior B with A.

Key point: Form sequential likelihoods before applying Bayes.

Q37 / D Identical letters constrained as block

Answer: (D) 840.

Given: MISSISSIPPI has M^1 I^4 S^4 P^2. Treat I^4 as one block.

Formula/Principle: Distinct permutations of multiset with eight objects equal 8!/(4!2!).

Step 1: The objects are block, M, four S and two P: total 8.

Step 2: 40320/(24×2)=840.

Worked derivation:

$$\text{objects} = \{I^4, M, S^4, P^2\} \text{ with } I^4 \text{ one block}; \quad N = \frac{8!}{4!\,2!} = 840.$$

Why: I's within their contiguous block are identical, so there is no internal 4! factor. The internal order of identical I's contributes only one arrangement. There are eight external objects after blocking, with repeated S and P requiring separate divisors.

Why not the others: (A) fails to divide by P repetition; (B) divides twice; (C) overcounts internal I swaps.

Key point: Count the block once and correct only remaining identical objects.

Q38 / C Adjacent binomial coefficients

Answer: (C) 7.

Given: Adjacent coefficients C(n,3) and C(n,4) are equal.

Formula/Principle: C(n,r+1)/C(n,r)=(n−r)/(r+1).

Step 1: Set (n−3)/4=1.

Step 2: n=7; indeed C(7,3)=C(7,4)=35.

Worked derivation:

$$\frac{\binom{n}{4}}{\binom{n}{3}} = \frac{n-3}{4} = 1 \Longrightarrow n = 7, \qquad \binom{7}{3} = \binom{7}{4} = 35.$$

Why: The row's two central adjacent coefficients tie only for odd n. Equating adjacent coefficients is a ratio condition; the equality places them symmetrically at the centre of an odd-index row.

Why not the others: (A), (B) and (D) make the ratio different from 1.

Key point: Use the ratio rather than expanding factorials.

Q39 / A Term from polynomial partial sum

Answer: (A) 29.

Given: $a_n=S_n-S_{n-1}$.

Formula/Principle: A partial sum includes all earlier terms, so adjacent sums differ by the new term.

Step 1: $S_5=3\times25+10=85$.

Step 2: $S_4=3\times16+8=56$; $a_5=85-56=29$.

Worked derivation:

$$a_5 = S_5 - S_4 = [3(5)^2 + 2(5)] - [3(4)^2 + 2(4)] = 85 - 56 = 29.$$

Why: Substituting n=5 directly gives a sum, not its fifth summand. A partial sum accumulates earlier terms. Taking the difference of adjacent sums isolates the fifth term without assuming the series is arithmetic.

Why not the others: (B), (C) and (D) follow incorrect index or algebra substitutions.

Key point: Difference consecutive partial sums.

Q40 / D Parameter cancellation in exponential limit

Answer: (D) (2,8).

Given: Denominator begins at cubic order $x^3/6$.

Formula/Principle: Compare leading Taylor terms after cancelling lower powers.

Step 1: Numerator quadratic coefficient is $2-a$; finite L requires $a=2$.

Step 2: Its cubic term is $(2x)^3/6=8x^3/6$, divided by $x^3/6$ gives $L=8$.

Worked derivation:

$$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + O(x^4), \quad e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + O(x^4); \quad a = 2,\ L = \frac{4/3}{1/6} = 8.$$

Why: A nonzero quadratic residual would make the quotient diverge like 1/x. A finite two-sided quotient requires the numerator's x^2 term to vanish because the denominator starts at x^3. The remaining leading coefficients then determine the limit.

Why not the others: (A) and (C) choose an uncancelled quadratic; (B) misses the cubing of 2.

Key point: Match the first nonzero orders before taking the ratio.

Q41 / B Eliminate geometric-family constants

Answer: (B) $yy''+(y')^2=0$.

Given: Differentiate $y^2=4a(x+b)$ once to get $2yy'=4a$, a constant.

Formula/Principle: Differentiate again to eliminate the constant a; b already disappears after the first derivative.

Step 1: Differentiate $2yy'=4a$ to get $2[(y')^2+yy'']=0$.

Step 2: Divide by 2 to obtain $yy''+(y')^2=0$.

Worked derivation:

$$y^2 = 4a(x+b) \Longrightarrow 2yy' = 4a \Longrightarrow 2(y')^2 + 2yy'' = 0 \Longrightarrow (y')^2 + yy'' = 0.$$

Why: Two arbitrary constants require a second-order relation; the product rule creates the squared slope term. The first derivative eliminates the shift b and the second eliminates the constant slope factor a. The second differentiation needs the product rule on yy'.

Why not the others: (A) uses a wrong product-rule sign; (C) drops curvature; (D) would force constant or zero branches.

Key point: Track how each differentiation removes a parameter.

Q42 / C Angle of line with plane

Answer: (C) 2/3.

Given: Plane normal n=(0,0,1) and line direction d=(1,2,2), |d|=3.

Formula/Principle: Sine of line-plane angle is |d·n|/(|d||n|).

Step 1: |d·n|=2.

Step 2: Divide by 3 to get 2/3.

Worked derivation:

$$\hat{\mathbf{n}} = (0,0,1), \quad |\mathbf{d}| = \sqrt{1+4+4} = 3, \quad \sin\theta = \frac{|\mathbf{d}\cdot\hat{\mathbf{n}}|}{|\mathbf{d}|} = \frac{2}{3}.$$

Why: This angle is complementary to the line-normal angle. The line-plane angle is complementary to the angle with the plane normal. The normal component, not the in-plane component, supplies its sine.

Why not the others: (A) uses x component; (B) uses in-plane projection; (D) assumes line normal to plane.

Key point: Use the component perpendicular to the plane.

Q43 / D Collinearity and internal division in three dimensions

Answer: (D) (3, 1:1).

Given: AC=(4,4,4) and AB=(a−1,2,2).

Formula/Principle: A point B on segment AC has AB=t AC with 0<t<1; its segment ratio is t:(1−t).

Step 1: Comparing y components gives 2=4t, so t=1/2.

Step 2: Comparing x gives a−1=4(1/2)=2, hence a=3; ratio is (1/2):(1/2)=1:1.

Worked derivation:

$$\overrightarrow{AC} = (4,4,4), \quad \overrightarrow{AB} = (a-1,2,2) = t\overrightarrow{AC}; \quad t = \frac{1}{2},\ a-1 = 2, \quad AB : BC = t : (1-t) = 1 : 1.$$

Why: The between condition validates the positive internal parameter instead of an extension. The y and z components independently confirm the same internal parameter. Since 0<t<1, the result satisfies the stipulated between condition.

Why not the others: (A) and (C) violate x collinearity; (B) has the correct x but wrong segment ratio.

Key point: Recover the common parameter and then the division ratio.

Q44 / A Mean absolute deviation from grouped frequencies

Answer: (A) 1.

Given: Total count=4 and weighted sum=0+2×2+4=8.

Formula/Principle: Mean deviation about mean is Σf|x−x̄|/Σf.

Step 1: x̄=8/4=2.

Step 2: Weighted absolute deviations=1×2+2×0+1×2=4; divide by 4 to get 1.

Worked derivation:

$$\bar{x} = \frac{1(0)+2(2)+1(4)}{4} = 2, \qquad \mathrm{MD}_{\bar{x}} = \frac{1|0-2|+2|2-2|+1|4-2|}{4} = 1.$$

Why: Deviations are absolute, so opposite signs do not cancel. Absolute deviations cannot cancel by sign, and the denominator is the total frequency four rather than the three distinct values.

Why not the others: (B) forgets frequency division; (C) divides twice; (D) uses the number of distinct values as denominator.

Key point: Use total observation count, not class count.

Q45 / C Quadratic reduction and periodic root count

Answer: (C) 6.

Given: Let s=sin x and use cos2x=1−2s².

Formula/Principle: Solve the quadratic condition in s, then count the valid angular branches over two periods.

Step 1: 2s²+s−1=0 gives s=1/2 or s=−1.

Step 2: On [0,4π], sin x=1/2 has four solutions and sin x=−1 has two; none coincide, total six.

Worked derivation:

$$s = \sin x, \quad s = 1 - 2s^2 \Longrightarrow (2s-1)(s+1) = 0; \quad \#\{s = 1/2\} = 4, \ \#\{s = -1\} = 2, \quad N = 6.$$

Why: Neither endpoint is a solution, while the second root is a tangent-level sine value occurring once each period. Two full periods lie in the interval. The level −1 occurs once per period while +1/2 occurs twice; neither endpoint adds an extra root.

Why not the others: (A) omits the −1 branch; (B) misses one period; (D) counts −1 twice per period.

Key point: Count each allowed sine level with its own multiplicity.

Physics

Q46 / A Piecewise acceleration and displacement

Answer: (A) 12 m.

Given: First-stage final speed=0+2×2=4 m/s.

Formula/Principle: Displacement in each constant-acceleration stage is $v_i t + at^2/2$.

Step 1: First displacement=0+2×4/2=4 m.

Step 2: Second displacement=4×4−1×16/2=8 m; total=12 m.

Worked derivation:

$$v_1 = 0 + (2)(2) = 4\,\text{m/s}, \quad s_1 = \tfrac{1}{2}(2)(2)^2 = 4\,\text{m}; \quad s_2 = (4)(4) + \tfrac{1}{2}(-1)(4)^2 = 8\,\text{m}; \quad s = 12\,\text{m}.$$

Why: Velocity stays nonnegative and reaches zero only at the end, so there is no reverse displacement. Velocity at the stage boundary becomes the initial velocity for the second stage. The final velocity is zero, so no reverse-motion correction is needed.

Why not the others: (B) counts only stage two; (C) only stage one; (D) omits braking loss.

Key point: Carry final velocity from one stage into the next.

Q47 / B Elevator apparent weight under acceleration reversal

Answer: (B) 480 N.

Given: True weight mg=600 N; observed upward-acceleration normal force=720 N.

Formula/Principle: N=m(g+a) upward and N=m(g−a) downward.

Step 1: a=720/60−10=2 m/s² upward initially.

Step 2: After reversal N=60(10−2)=480 N.

Worked derivation:

$$N_1 = m(g+a) = 720 \Rightarrow a = \frac{720}{60} - 10 = 2\,\mathrm{m/s^2}; \quad N_2 = m(g-a) = 60(10-2) = 480\,\mathrm{N}.$$

Why: The scale reports contact force, which depends on acceleration, not velocity direction. The scale measures normal force, not mg alone. Reversing acceleration at the same magnitude switches the sign of ma regardless of the elevator's velocity.

Why not the others: (A) assumes zero acceleration; (C) ignores reversal; (D) doubles the inferred acceleration.

Key point: Infer acceleration from the first reading before changing its sign.

Q48 / C Energy with dissipative horizontal stage

Answer: (C) 60 m²/s².

Worked derivation:

$$\tfrac{1}{2}v^2 = gh - \mu_k g d = (10)(5) - (0.20)(10)(10) = 30\,\mathrm{m^2/s^2}, \qquad v^2 = 60\,\mathrm{m^2/s^2}.$$

Why: Descent provides mgh=50m joules; friction on the horizontal segment removes μmgd=.2×m×10×10=20m joules. The remaining kinetic energy is 30m=(1/2)mv², so v²=60. The sloping segment is frictionless, whereas the shaded horizontal segment dissipates energy. Compute the available height energy and the rough-track loss separately before solving for v.

Why not the others: (A) ignores friction; (B) removes too little; (D) treats loss as final energy.

Key point: Locate where friction acts and conserve energy with its work included.

Q49 / D Angular acceleration with torque and inertia change

Answer: (D) It remains unchanged.

Given: τ′=2τ and I′=2I.

Formula/Principle: τ=Iα about a fixed axis.

Step 1: α′=τ′/I′=2τ/(2I).

Step 2: α′=α.

Worked derivation:

$$\alpha = \frac{\tau}{I}, \quad \alpha' = \frac{2\tau}{2I} = \alpha.$$

Why: Torque and rotational inertia scale together; their ratio sets acceleration. Torque and inertia change together. The rotational form of Newton's second law depends on their ratio, so doubling both does not double acceleration.

Why not the others: (A) multiplies both changes; (B) changes only torque; (C) changes only inertia.

Key point: Compare the ratio, not either factor alone.

Q50 / A Gravity within uniform sphere

Answer: (A) g(R)/2.

Given: Enclosed mass at radius r scales as r³ for uniform density.

Formula/Principle: g(r)=G M(r)/r², hence g(r)∝r inside the sphere.

Step 1: M(R/2)=M(R)/8.

Step 2: Divide by (R/2)²=R²/4, yielding (1/8)/(1/4)=1/2 of surface g.

Worked derivation:

$$M(r) = M(R)\left(\frac{r}{R}\right)^3, \quad g(r) = \frac{GM(r)}{r^2} = g(R)\frac{r}{R}; \qquad g(R/2) = \frac{1}{2}g(R).$$

Why: Outer spherical shells exert zero net field at an internal point. Inside a uniform sphere only enclosed mass contributes net field. Using the full mass at r=R/2 would incorrectly apply the exterior inverse-square law.

Why not the others: (B) uses only inverse-square distance; (C) and (D) ignore enclosed-mass reduction.

Key point: Use enclosed mass, not total planetary mass, inside.

Q51 / B Stokes terminal speed scaling

Answer: (B) 4.

Given: Material densities, fluid viscosity and g are shared.

Formula/Principle: Balancing effective weight ∝r³ with Stokes drag ∝rv gives terminal speed ∝r².

Step 1: Radius ratio=2.

Step 2: Speed ratio=2²=4.

Worked derivation:

$$(\rho_s - \rho_f)\frac{4}{3}\pi r^3 g = 6\pi\eta r v_t \Longrightarrow v_t = \frac{2(\rho_s - \rho_f)gr^2}{9\eta}, \quad \frac{v_t(2r)}{v_t(r)} = 4.$$

Why: Buoyancy changes the density difference but not the radius exponent under the stated conditions. This scaling assumes Stokes flow and equal material and fluid properties. Buoyancy changes the density difference but cancels in the ratio.

Why not the others: (A) uses linear scaling; (C) uses volume scaling; (D) squares twice.

Key point: Derive scaling from force balance.

Q52 / C Adiabatic ideal-gas temperature ratio

Answer: (C) 2^(−2/3).

Given: Vf/Vi=2 and γ−1=2/3.

Formula/Principle: For reversible adiabatic ideal-gas change, TV^(γ−1)=constant.

Step 1: Tf/Ti=(Vi/Vf)^(γ−1).

Step 2: Substitute (1/2)^(2/3)=2^(−2/3).

Worked derivation:

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}, \quad \frac{T_f}{T_i} = \left(\frac{V_i}{2V_i}\right)^{5/3-1} = 2^{-2/3}.$$

Why: The gas cools on adiabatic expansion because it does work without heat input. The process is reversible adiabatic, which is what licenses this power law. Expansion lowers temperature because internal energy supplies the work.

Why not the others: (A) assumes exponent one; (B) predicts heating; (D) uses γ instead of γ−1.

Key point: Use the temperature-volume exponent γ−1.

Q53 / D Flux with enclosed charges

Answer: (D) 2q/ε₀.

Given: Algebraic enclosed charge q−q+2q=2q.

Formula/Principle: Gauss's law ∮E·dA=Q_enclosed/ε₀.

Step 1: Cancel the +q and −q contributions in charge sum.

Step 2: Remaining +2q gives positive outward flux 2q/ε₀.

Worked derivation:

$$Q_{\text{enc}} = q - q + 2q = 2q, \qquad \Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = \frac{2q}{\varepsilon_0}.$$

Why: External charges could reshape the local field but cannot change net closed-surface flux. Gauss's law uses algebraic enclosed charge. External charges and uneven local field strength may alter the pattern on the surface but not its net flux.

Why not the others: (A) cancels all charges; (B) misses a q; (C) adds magnitudes without signs.

Key point: Use algebraic enclosed charge.

Q54 / A Series capacitors and shared charge

Answer: (A) 2CV/3.

Given: Series capacitances are C and 2C.

Formula/Principle: Equivalent capacitance is C_eq=(C×2C)/(C+2C)=2C/3; both series elements carry equal charge magnitude.

Step 1: Total series charge Q=C_eq V.

Step 2: Q=2CV/3 on either capacitor.

Worked derivation:

$$C_{\text{eq}} = \left(\frac{1}{C} + \frac{1}{2C}\right)^{-1} = \frac{2C}{3}, \qquad Q_{\text{series}} = C_{\text{eq}} V = \frac{2CV}{3}.$$

Why: Unequal capacitance creates unequal voltage drops, not unequal series charge. In a simple series chain each capacitor has the same magnitude of charge, even though the voltage across the two capacitances differs.

Why not the others: (B) halves the shared charge; (C) and (D) apply total V directly to an individual capacitor.

Key point: Calculate series equivalent capacitance before charge.

Q55 / B Magnetic dipole potential energy

Answer: (B) +pB/2.

Worked derivation:

$$U_i = -pB\cos 0^\circ = -pB, \quad U_f = -pB\cos 60^\circ = -\tfrac{1}{2}pB; \qquad W_{\text{ext}} = \Delta U = \tfrac{1}{2}pB.$$

Why: Magnetic potential energy is U=−pBcosθ. It changes from −pB at 0° to −pB/2 at 60°, an increase of pB/2 supplied by an external agent in a slow rotation. The quasistatic qualifier lets external work equal the energy increase without a change in kinetic energy; magnetic field work has the opposite sign.

Why not the others: (A) uses field work rather than external work; (C) treats the final energy as zero; (D) ignores orientation dependence.

Key point: Quasistatic external work equals the change in potential energy.

Q56 / C Mutual induction signed emf

Answer: (C) −1 V.

Given: M=0.50 H and dI_1/dt=+2 A/s.

Formula/Principle: Signed mutual emf $\mathcal{E}_2$=−M dI_1/dt under matching orientation convention.

Step 1: Multiply 0.50×2=1 V.

Step 2: Lenz's minus sign gives −1 V.

Worked derivation:

$$\mathcal{E}_2 = -M\frac{dI_1}{dt} = -(0.50\,\mathrm{H})(2\,\mathrm{A/s}) = -1\,\mathrm{V}.$$

Why: The secondary opposes the increase in linked positive flux; the sign depends on the explicitly chosen convention. The given matching positive flux convention fixes the negative sign. Without an orientation convention only the magnitude could be reported unambiguously.

Why not the others: (A) drops Lenz's sign; (B) misses the factor M; (D) misses both.

Key point: State a flux orientation when interpreting emf sign.

Q57 / D Lens-plane mirror collimation condition

Answer: (D) 12.5 cm.

Given: First-pass lens image is 15 cm right of lens by 1/f=1/30+1/v.

Formula/Principle: Plane mirror at d reflects a converging point at x=15 into a return-side point at x=2d−15; reverse lens output is parallel if that point is at its 10 cm focal plane.

Step 1: Set 2d−15=10.

Step 2: d=12.5 cm, indeed before x=15 as specified.

Worked derivation:

$$\frac{1}{10} = \frac{1}{30} + \frac{1}{v_1} \Rightarrow v_1 = 15\,\mathrm{cm}; \quad x_{\text{return}} = 2d - 15; \quad x_{\text{return}} = f = 10 \Rightarrow d = 12.5\,\mathrm{cm}.$$

Why: The mirror folds the convergent bundle; its image point supplies the object location for the second lens pass. The first image is virtual to the mirror because it lies behind it at d<15. Reflecting that convergent bundle produces a return-side point at 2d−15; placing that point at the lens focus yields parallel output.

Why not the others: (A), (B), (C) put the return point at −5,5,15 cm, not 10.

Key point: Track both lens traversals and the reflected intermediate point.

Q58 / D Fringe visibility from intensities

Answer: (D) 4/5.

Worked derivation:

$$\mathcal{V} = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}} = \frac{9-1}{9+1} = \frac{4}{5}.$$

Why: Visibility is (Imax−Imin)/(Imax+Imin), which here is (9−1)/(9+1)=8/10=4/5. A nonzero minimum indicates imperfect contrast, so visibility is below one. The nonzero minimum prevents unit visibility. The sum in the denominator normalises contrast against total intensity rather than using a raw 8-unit difference.

Why not the others: (A) ignores the intensity offset; (B) requires zero minimum; (C) uses the minimum-to-maximum ratio incorrectly.

Key point: Use both extrema in the normalised contrast.

Q59 / B Hydrogen series transition from wavelength

Answer: (B) n=4.

Given: Final level is 2 and spectral wavenumber is 3R/16.

Formula/Principle: $1/\lambda = R(1/2^2 - 1/n_i^2)$.

Step 1: $1/4 - 1/n_i^2 = 3/16$, so $1/n_i^2 = 1/16$.

Step 2: Positive principal quantum number $n_i = 4$.

Worked derivation:

$$\tilde{\nu} = R\left(\frac{1}{2^2} - \frac{1}{n_i^2}\right) = \frac{3R}{16} \Rightarrow \frac{1}{n_i^2} = \frac{1}{16} \Rightarrow n_i = 4.$$

Why: Energy difference, not ratio of level numbers, determines photon wavenumber. The initial level must exceed the terminating level two for emission; the positive integer solution four satisfies that physical direction.

Why not the others: (A), (C), (D) give different inverse-square differences.

Key point: Rearrange the Rydberg equation before extracting the level.

Q60 / C Activity decay in a mixture

Answer: (C) 1:2.

Given: Initial activity A_0 is equal; elapsed time is 4 h.

Formula/Principle: A(t)=A_0(1/2)^(t/T_half) for each isotope separately.

Step 1: First decays over two half-lives to $A_0/4$.

Step 2: Second over one half-life to $A_0/2$, ratio=(1/4):(1/2)=1:2.

Worked derivation:

$$A_1(4\,\mathrm{h}) = A_0\,2^{-4/2} = A_0/4, \qquad A_2(4\,\mathrm{h}) = A_0\,2^{-4/4} = A_0/2, \quad A_1 : A_2 = 1 : 2.$$

Why: Equal starting activity does not imply equal remaining activity with different decay constants. Equal starting activities do not mean equal initial nuclei when half-lives differ. Each activity decays by its own half-life over the same elapsed time.

Why not the others: (A) squares only one ratio; (B) reverses order; (D) ignores elapsed time.

Key point: Compare fractions after the same clock time.
